

Project Proposal ↗

Accident Risk & Severity
Prediction

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Summary

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This project proposes a data-driven approach to analyze and predict traffic accident risks using historical US accident data. By applying machine learning models and spatial analysis techniques, the system identifies accident hotspots, predicts accident severity, and visualizes risk patterns. The solution aims to support transportation authorities in improving road safety and implementing preventive measures.

Problem Statement

01

Road accidents cause injuries, deaths, and economic losses.

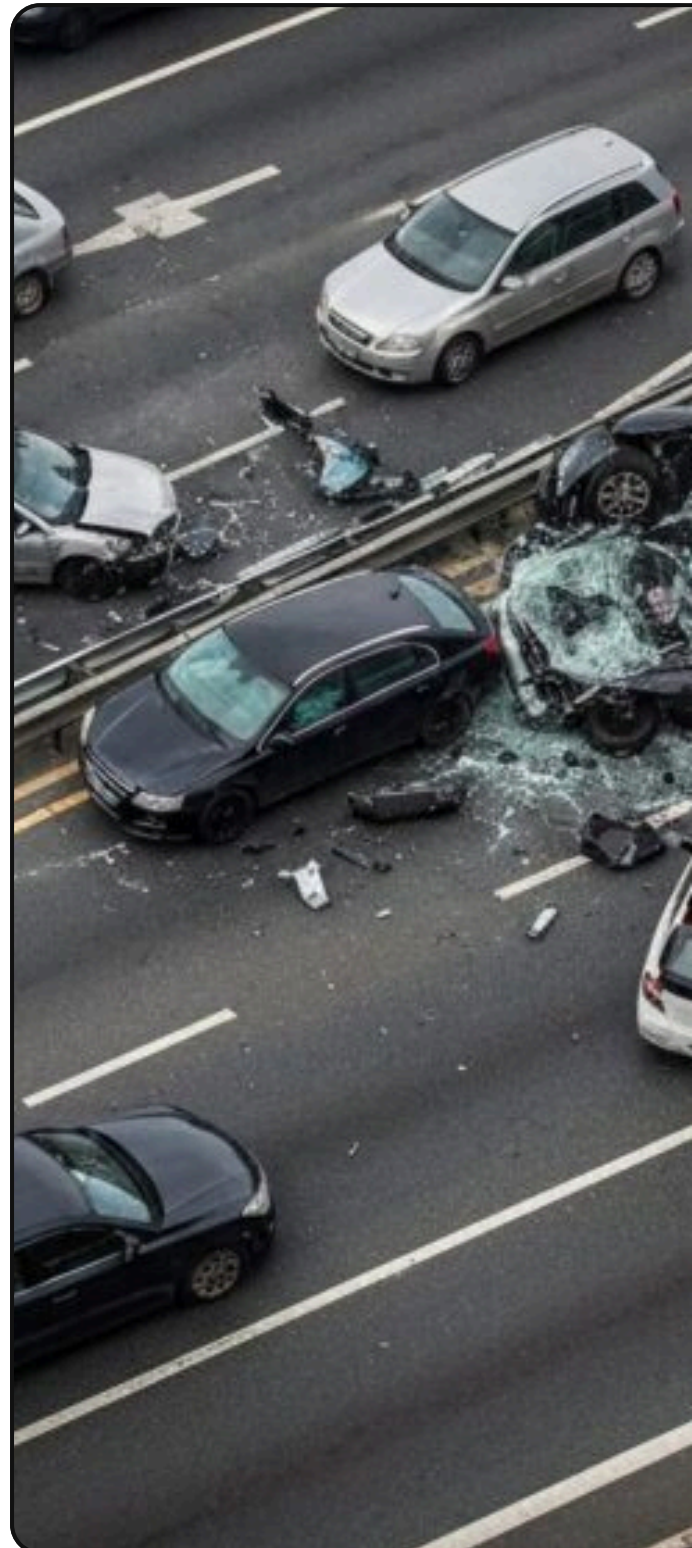
02

Authorities often rely on historical reports rather than predictive analytics.

03

Limited understanding of accident patterns hinders proactive prevention measures.

There is a need for a system that transforms raw accident data into actionable insights to improve road safety and reduce accident risks.



Project Objectives

➔ Key Objectives

- Analyze historical US accident data to identify patterns and trends.
- Detect high-risk accident hotspots using clustering techniques.
- Predict accident severity levels using machine learning models.
- Visualize spatial accident distribution through interactive maps and dashboards.
- Provide insights for policy making and road safety planning.

Data Sources and Description

➔ Key Deliverables

- The analysis is based on the US Accidents Dataset from Kaggle, which contains more than 7 millions of accident records across the United States. The dataset includes information on location, weather conditions, road infrastructure, and accident severity, enabling pattern analysis and predictive modeling.
- This extensive dataset enables the analysis of accident patterns, identification of high-risk areas, and development of predictive models for accident severity.



Data Sources and Description

Key Features in the Dataset

Feature	Description
Start_Lat / Start_Lng	Geographic coordinates indicating the exact location of the accident.
Severity	The severity level of the accident, typically categorized into multiple levels.
Weather_Condition	Weather conditions at the time of the accident (e.g., rain, fog, clear).
Visibility	The visibility distance at the accident location.
Temperature	Temperature recorded at the time of the accident.
Traffic_Signal	Indicates the presence of traffic signals near the accident location.
Road_Conditions	Information about road characteristics and environmental factors that may influence accident occurrence.

Proposed Solution

01

Exploratory Data Analysis

Identify patterns in accident frequency.

02

Spatial Risk Mapping

Visualize accident hotspots using geographic coordinates.

03

Clustering Analysis

Use DBSCAN or K-Means to detect accident clusters.

04

Machine Learning Prediction

Predict accident risk score and severity level.

05

Interactive Dashboard

Display insights and risk zones.

Methodology and Technical Approach

➔ Data Preprocessing

Handle missing values , Feature selection , Feature encoding , Normalization

➔ Exploratory Data Analysis (EDA)

Accident distribution by time, weather , location , road conditions

➔ Feature Engineering

Time-based features (hour, day, weekend) , Weather risk indicators , Traffic infrastructure indicators

➔ Machine Learning Models

predict accident severity using multi-class classification and estimate accident risk scores.

Clustering and geospatial mapping will also be applied to identify and visualize high-risk accident hotspots.

➔ Model Evaluation

Accuracy , Precision , Recall , F1 Score

Expected Outcomes

➔ Key Objectives

- Identify high-risk locations
- Detect high-risk accident hotspots using clustering techniques.
- Predict severity of potential accidents
- Improve traffic planning
- Assist government safety policies
- Reduce accident rates

Tools and Technologies

Category	Tools
Programming	Python
Data Processing	Pandas, NumPy
Visualization	Seaborn, Matplotlib
Spatial Analysis	Folium
Machine Learning	Scikit-learn
Dashboard	Power BI
Version Control	Git, GitHub

Risk Assessment and Limitations

01

Missing data in weather or road conditions

02

Model accuracy depends on historical patterns

03

Geographic bias in dataset coverage

04

Lack of Real-Time Information

05

Driver Behavior Data





Timeline and Milestones

Phase	Duration
Data Preparation	1 week
EDA	1 week
Model Development	2 weeks
Visualization	1 week
Final Report	1 week

Future Enhancements

Integrate real-time traffic data

Use deep learning models

Develop live traffic risk monitoring system

Integrate Driver Behavior Data



Conclusion



This project demonstrates how data analytics and machine learning can transform accident data into actionable insights.

By identifying risk zones and predicting accident severity, the proposed solution can support data-driven decision-making for safer transportation systems.

Let's Work Together ↗

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Pipeline

➔ Data Collection

Handle missing values , Feature selection , Feature encoding , Normalization

➔ Data Cleaning

Accident distribution by time, weather , location , road conditions

➔ Exploratory Data Analysis

Time-based features (hour, day, weekend) , Weather risk indicators , Traffic infrastructure indicators

➔ Feature Engineering

Time-based features (hour, day, weekend) , Weather risk indicators , Traffic infrastructure indicators

➔ Model Training

Accuracy , Precision , Recall , F1 Score

➔ Model Evaluation

Accuracy , Precision , Recall , F1 Score

Future Enhancements

Risk	Likelihood	Impact	Mitigation Plan
Delayed content creation	Medium	High	Develop content calendar
Ad campaign underperformance	Medium	Medium	Conduct A/B testing
Budget overruns	Low	High	Regular budget tracking
Platform algorithm changes	High	Medium	Stay updated with trends





Conclusion

01

Increased website traffic and improved search rankings.

02

Enhanced social media presence and engagement.

03

Higher lead generation and conversion rates.

04

Strengthened brand reputation and visibility.

05

Actionable insights from analytics to guide future strategies.